

Bluestock Fintech — Mutual Fund Analytics Platform

Final Capstone Report | June 2026

1. Executive Summary

This report presents the findings of a one-week capstone project undertaken for Bluestock Fintech. The objective was to design and build a complete Mutual Fund Analytics Platform covering data ingestion, ETL pipeline, exploratory data analysis, performance metrics computation, and interactive dashboard development.

The platform analyses 40 mutual fund schemes across 10 categories, covering 4.5 years of daily NAV data, 55,000+ investor transactions, and quarterly AUM data for the top 10 fund houses in India.

Key findings include SBI Mutual Fund's dominant position with Rs.12.5 lakh crore in AUM, industry SIP inflows reaching an all-time high of Rs.31,002 crore in December 2025, and significant quality variation across funds with Sharpe ratios ranging from negative to above 1.5.

2. Problem Statement

The Indian mutual fund industry manages over Rs.81 lakh crore across 26 crore investor folios as of December 2025. Despite this scale, retail investors face five core challenges: fragmented data across multiple sources, no easy way to compare funds on a risk-adjusted basis, no benchmark tracking, limited visibility into investor behaviour, and slow static reporting.

This platform addresses all five problems through a unified data pipeline and interactive dashboard.

3. Data Sources and Datasets

Ten datasets were used covering fund metadata, daily NAV history, AUM figures, SIP inflows, investor transactions, portfolio holdings, and benchmark indices. All data is sourced from publicly available AMFI India publications, mfapi.in, and NSE/BSE open data.

The NAV history dataset contains approximately 46,000 rows of daily prices anchored to real AMFI values. The investor transaction dataset contains 55,000+ records across 5,000 investors in 12 Indian states.

4. System Architecture

The platform follows a five-layer architecture:

Layer 1 — Extract: Raw CSV datasets loaded via Python and live NAV data fetched from the mfapi.in REST API.

Layer 2 — Transform: Pandas-based cleaning pipeline handling missing values, date parsing, duplicate removal, and forward-filling of NAV data for weekends and holidays.

Layer 3 — Load: All cleaned data loaded into a SQLite database with a 5-table star schema centred on dim_fund.

Layer 4 — Analyse: Jupyter notebooks computing 15+ EDA charts and all key risk-return metrics including CAGR, Sharpe, Sortino, Alpha, Beta, VaR, CVaR, and HHI.

Layer 5 — Visualise: Power BI dashboard with 4 pages, slicers, and drill-through capability.

5. EDA Findings

Monthly SIP inflows grew consistently from Rs.11,000 crore in January 2022 to Rs.31,002 crore in December 2025 reflecting deepening retail participation. SBI Mutual Fund maintained market leadership throughout with AUM growing from approximately Rs.6 lakh crore to Rs.12.5 lakh crore over the same period.

The 26-35 age group accounts for the largest share of SIP transactions while T30 cities contribute approximately 65% of total investment volume. Financials and IT sectors together represent over 45% of equity fund portfolio weight.

6. Performance Analytics

CAGR analysis shows Small Cap funds delivering the highest 5-year returns but also the highest volatility. Large Cap funds show more consistent risk-adjusted returns with Sharpe ratios averaging above 0.8. The fund composite scorecard combines five metrics — 3yr CAGR, Sharpe ratio, Alpha, expense ratio and Max Drawdown — into a single 0-100 score for easy comparison.

Benchmark comparison reveals that most actively managed Large Cap funds have struggled to consistently outperform the Nifty 100 index over the 3-year period, supporting the case for index fund investing for that category.

7. Advanced Analytics

Value at Risk analysis confirms that Small Cap funds carry the highest daily risk with 95% VaR exceeding -2% while Liquid funds remain near zero. Rolling Sharpe ratio analysis reveals that even top-performing funds experience extended periods of below-average risk-adjusted returns.

Investor cohort analysis shows that 2024 cohort investors begin with higher average SIP amounts than 2022 cohorts. SIP continuation analysis identified approximately 30% of active investors as at-risk based on irregular payment gaps exceeding 35 days.

8. Dashboard

The Power BI dashboard delivers four interactive pages covering industry overview, fund performance, investor analytics, and market trends. Every page includes at least two slicers enabling self-service exploration without technical knowledge.

9. Recommendations

Based on the analysis, three recommendations are made for Bluestock Fintech. First, prioritise retention campaigns targeting at-risk SIP investors in Maharashtra and Delhi where the highest concentration of irregular investors was found. Second, promote index funds to Large Cap investors given the evidence that

most active Large Cap funds underperform Nifty 100 net of expenses. Third, develop targeted products for the 2024 cohort which shows higher income levels and larger initial SIP commitments.

10. Limitations

The investor transaction dataset is synthetically generated using real demographic distributions but does not represent actual investor records. NAV data beyond real anchor points is simulated using geometric Brownian motion with realistic parameters. Results should be interpreted in the context of this educational project.

11. Conclusion

This capstone successfully demonstrates an end-to-end data engineering and analytics workflow applied to the Indian mutual fund domain. All eight project objectives were met within the seven-day timeline.

Prepared by Anish Abhyankar | Bluestock Fintech Data Analyst Internship | June 2026